# Word Problems & Reasoning

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Number and digit reasoning**

A number decreased by 9 equals 34. Find the number.

$x - 9 = 34 \Rightarrow x = 43$.

**2** **[Foundational]** • **Target: Number and digit reasoning**

Three times a number, minus 7, equals 32. Find the number.

$3x - 7 = 32 \Rightarrow 3x = 39 \Rightarrow x = 13$.

**3** **[Foundational]** • **Target: Number and digit reasoning**

The sum of two numbers is 58. One number is 12 more than the other. Find both numbers.

Let the smaller be $x$: $x + (x + 12) = 58 \Rightarrow 2x = 46 \Rightarrow x = 23$. The numbers are $23$ and $35$.

**4** **[Foundational]** • **Target: Number and digit reasoning**

A two-digit number has digit sum 11. Reversing its digits gives a number 45 more than the original. Find the original number.

Let the digits be $t$ (tens) and $u$ (units): $t + u = 11$ and $(10u + t) - (10t + u) = 45 \Rightarrow u - t = 5$. Solving together: $u = 8$, $t = 3$. The original number is $38$.

**5** **[Foundational]** • **Target: Age problems**

A father is 4 times as old as his daughter. In 5 years, he will be 3 times as old as she will be. Find their current ages.

Let the daughter be $x$, father $4x$: $4x + 5 = 3(x + 5) \Rightarrow 4x + 5 = 3x + 15 \Rightarrow x = 10$. Daughter is $10$, father is $40$.

**6** **[Foundational]** • **Target: Age problems**

Sara is twice as old as her brother. In 6 years, the sum of their ages will be 42. Find their current ages.

Let the brother be $x$, Sara $2x$: $(x + 6) + (2x + 6) = 42 \Rightarrow 3x = 24 \Rightarrow x = 10$. Brother is $10$, Sara is $20$.

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**7** [Foundational] • Target: Age problems

Ahmed is 5 years older than Salem. Seven years ago, Ahmed was twice as old as Salem. Find their current ages.

Let Salem be $x$, Ahmed $x+5$: $(x+5-7)=2(x-7) \Rightarrow x-2=2x-14 \Rightarrow x=12$. Salem is $12$, Ahmed is $17$.

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Age problems

The sum of a mother's and daughter's ages is 50. The mother is 20 years older than the daughter. Find both ages.

Let the daughter be $d$: $d+(d+20)=50 \Rightarrow 2d=30 \Rightarrow d=15$. Daughter is $15$, mother is $35$.

**9** [Intermediate] • Target: Distance, rate and time

A train travels 180 km in 2 hours. At the same speed, how far does it travel in 5 hours?

Speed $=180 \div 2=90$ km/h. Distance $=90 \times 5=450$ km.

**10** [Intermediate] • Target: Distance, rate and time

Two cities are 350 km apart. A car leaves one city for the other at 70 km/h. How long does the trip take?

Time $=350 \div 70=5$ hours.

**11** [Intermediate] • Target: Distance, rate and time

Two cyclists start at the same point and ride in opposite directions, at 15 km/h and 20 km/h. After how many hours will they be 105 km apart?

Combined speed $=15+20=35$ km/h. Time $=105 \div 35=3$ hours.

**12** [Intermediate] • Target: Distance, rate and time

A plane flies against a headwind at 480 km/h and covers 1,440 km. How many hours does the flight take?

Time $=1{,}440 \div 480=3$ hours.

**13** [Intermediate] • Target: Work-rate problems

Worker A can paint a room in 6 hours; Worker B can paint the same room in 3 hours. Working together, how long will it take them?

Combined rate $=\frac{1}{6}+\frac{1}{3}=\frac{1}{6}+\frac{2}{6}=\frac{3}{6}=\frac{1}{2}$ room/hour. Time $=2$ hours.

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**14** [Intermediate] • Target: Work-rate problems

A pipe fills a tank in 10 hours; a second pipe fills it in 15 hours. Working together, how long to fill the tank?

Combined rate $= \frac{1}{10} + \frac{1}{15} = \frac{3}{30} + \frac{2}{30} = \frac{5}{30} = \frac{1}{6}$. Time $= 6$ hours.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Work-rate problems • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

6 workers can complete a job in 8 days. How many days would 4 workers take, at the same rate?

Total work $= 6 \times 8 = 48$ worker-days. With $4$ workers: $48 \div 4 = 12$ days.

**16** [Challenging] • Target: Work-rate problems • Cross-Lesson: 3.01 Algebra Essentials

A pipe fills a tank in 12 hours, but a leak can empty a full tank in 24 hours. If both act at once, how long to fill the tank?

Net rate $= \frac{1}{12} - \frac{1}{24} = \frac{2}{24} - \frac{1}{24} = \frac{1}{24}$. Time $= 24$ hours.

**17** [Challenging] • Target: Mixture and money problems • Cross-Lesson: 4.01 Geometry & Measurement

A jeweler mixes gold worth 200 riyals/gram with silver worth 50 riyals/gram to make 10 grams of a mixture worth 110 riyals/gram. How many grams of gold are used?

Let $g$ be grams of gold: $200g + 50(10 - g) = 110 \times 10 \Rightarrow 150g = 600 \Rightarrow g = 4$ grams.

**18** [Challenging] • Target: Mixture and money problems • Cross-Lesson: 5.01 Quantitative Comparisons

A shopkeeper has 15 riyals in 1-riyal and 2-riyal coins, 10 coins in total. How many of each coin does she have?

Let $x$ be 1-riyal coins, $y$ be 2-riyal coins: $x + y = 10$, $x + 2y = 15$. Subtracting: $y = 5$, $x = 5$.

**19** [Challenging] • Target: Mixture and money problems • Cross-Lesson: 6.01 Sequences & Patterns

A solution is $40\%$ acid. How many liters of pure water must be added to 20 liters of this solution to dilute it to $25\%$ acid?

Acid amount stays $0.4 \times 20 = 8$ liters. $\frac{8}{20 + w} = 0.25 \Rightarrow 20 + w = 32 \Rightarrow w = 12$ liters.

**20** [Challenging] • Target: Mixture and money problems • Cross-Lesson: 7.01 Data Analysis & Probability

Event tickets cost 30 riyals for adults and 18 riyals for children. 150 tickets were sold for 3,780 riyals in total. How many adult tickets were sold?

Let $a$ be adults, so $150 - a$ children: $30a + 18(150 - a) = 3{,}780 \Rightarrow 12a = 1{,}080 \Rightarrow a = 90$ adult tickets.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: MCQ — Number and digit reasoning

A number decreased by 12 equals 47. Find the number.

**a.** 59 ✓ **b.** 65

**c.** 35 **d.** 47

$x - 12 = 47 \Rightarrow x = 59$.

**22** [Foundational] • Target: MCQ — Number and digit reasoning

A two-digit number has digit sum 9. Reversing its digits gives a number 27 more than the original. Find the original number.

**a.** 18 **b.** 36 ✓

**c.** 63 **d.** 45

Digits $3, 6$: original $= 36$, reversed $= 63 = 36 + 27$.

**23** [Foundational] • Target: MCQ — Age problems

Huda is twice as old as her sister. In 8 years, the sum of their ages will be 46. Find Huda's current age.

**a.** 10 **b.** 18

**c.** 14 **d.** 20 ✓

$(x + 8) + (2x + 8) = 46 \Rightarrow x = 10$; Huda $= 20$.

**24** [Foundational] • Target: MCQ — Age problems

Tariq is currently twice as old as his son. If Tariq is 40, find his son's age.

**a.** 18 **b.** 20 ✓

**c.** 22 **d.** 80

$40 = 2s \Rightarrow s = 20$.

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**25** [Foundational] • Target: MCQ — Distance, rate and time

A plane flies against a headwind at 400 km/h and covers 1,600 km. How many hours does the flight take?

**a.** 4,000 **b.** 4 ✓

**c.** 5 **d.** 400

1,600 ÷ 400 = 4 hours.

**26** [Foundational] • Target: MCQ — Work-rate problems

8 workers can complete a job in 6 days. How many days would 12 workers take, at the same rate?

**a.** 4 ✓ **b.** 9

**c.** 3 **d.** 6

Total work = 48 worker-days; 48 ÷ 12 = 4 days.

**27** [Foundational] • Target: MCQ — Mixture and money problems

A shopkeeper has 21 riyals in 1-riyal and 2-riyal coins, 15 coins in total. How many 2-riyal coins does he have?

**a.** 9 **b.** 15

**c.** 6 ✓ **d.** 3

$x + y = 15, x + 2y = 21 \Rightarrow y = 6.$

**28** [Foundational] • Target: MCQ — Logical deduction

Omar, Bandar and Nasser have 44 books together. Omar has twice as many as Bandar, and Nasser has 8 more than Bandar. Find how many books Bandar has.

**a.** 22 **b.** 17

**c.** 9 ✓ **d.** 18

$b + 2b + (b + 8) = 44 \Rightarrow 4b = 36 \Rightarrow b = 9.$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: MCQ — Logical deduction

Three consecutive integers sum to 72. Find the smallest one.

**a.** 25

**b.** 24

**c.** 23 ✓

**d.** 22

$3n + 3 = 72 \Rightarrow n = 23$.

**30** [Intermediate] • Target: MCQ — Working backward

After a 20% pay cut, an employee's salary is 3,200 riyals. Find the salary before the cut.

**a.** 4,200

**b.** 4,000 ✓

**c.** 3,600

**d.** 3,840

$0.8x = 3{,}200 \Rightarrow x = 4{,}000$.

**31** [Intermediate] • Target: MCQ — Mixed word problems

Sarah buys 4 notebooks and 3 pens for 38 riyals. Khalid buys 2 notebooks and 5 pens for 40 riyals, at the same prices. Find the price of one notebook.

**a.** 6

**b.** 8

**c.** 5 ✓

**d.** 4

Solving the system: notebook $= 5$ riyals, pen $= 6$ riyals.

**32** [Intermediate] • Target: MCQ — Mixed word problems

A cyclist travels 60 km at a certain speed, then returns the same 60 km at half that speed, taking 9 hours total. Find the original speed.

**a.** 15

**b.** 10

**c.** 40

**d.** 20 ✓

$\frac{60}{v} + \frac{120}{v} = 9 \Rightarrow \frac{180}{v} = 9 \Rightarrow v = 20$ km/h.

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**33** [Intermediate] • Target: MCQ — More rate and proportion problems

A factory produces 1,500 items in 6 hours. At the same rate, how many items would it produce in a 9-hour shift?

**a.** 1,875

**b.** 2,700

**c.** 2,250 ✓

**d.** 2,000

Rate $= 250$/hr; $9 \times 250 = 2{,}250$.

**34** [Intermediate] • Target: MCQ — Work-rate problems

A number decreased by 15 equals 26. Find the number.

**a.** 41 ✓

**b.** 39

**c.** 46

**d.** 43

$x - 15 = 26 \Rightarrow x = 41$.

**35** [Intermediate] • Target: MCQ — Mixture and money problems

A number decreased by 13 equals 66. Find the number.

**a.** 79 ✓

**b.** 77

**c.** 84

**d.** 81

$x - 13 = 66 \Rightarrow x = 79$.

**36** [Intermediate] • Target: MCQ — Logical deduction

A number decreased by 38 equals -19. Find the number.

**a.** 21

**b.** 19 ✓

**c.** 17

**d.** 24

$x - 38 = -19 \Rightarrow x = 19$.

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**37** [Intermediate] • Target: MCQ — Working backward

Three times a number, minus 13, equals 17. Find the number.

**a.** 15

**b.** 8

**c.** 12

**d.** 10 ✓

$3x - 13 = 17 \Rightarrow 3x = 30 \Rightarrow x = 10$.

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: MCQ — Mixed word problems • Cross-Lesson: 5.01 Quantitative Comparisons

A train travels 535 km in 5 hours. At the same speed, how far does it travel in 2 hours?

**a.** 219

**b.** 212

**c.** 214 ✓

**d.** 216

Speed $= 535 \div 5 = 107$ km/h. Distance $= 107 \times 2 = 214$ km.

**39** [Challenging] • Target: MCQ — More rate and proportion problems • Cross-Lesson: 6.01 Sequences & Patterns

Two cities are 294 km apart. A car leaves one city for the other at 49 km/h. How long does the trip take?

**a.** 4

**b.** 8

**c.** 11

**d.** 6 ✓

Time $= 294 \div 49 = 6$ hours.

**40** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Data Analysis & Probability

A train travels 202 km in 2 hours. At the same speed, how far does it travel in 5 hours?

**a.** 510

**b.** 503

**c.** 507

**d.** 505 ✓

Speed $= 202 \div 2 = 101$ km/h. Distance $= 101 \times 5 = 505$ km.

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**41** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Arithmetic & Number Sense

A train travels 540 km in 6 hours. At the same speed, how far does it travel in 8 hours?

**a.** 720 ✓ **b.** 725

**c.** 718 **d.** 722

Speed $= 540 \div 6 = 90$ km/h. Distance $= 90 \times 8 = 720$ km.

**42** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

Yousef is 5 times as old as his younger sibling. In 6 years, the sum of their ages will be 60. Find their current ages.

**a.** 8, 40 ✓ **b.** 8, 44

**c.** 10, 40 **d.** 40, 8

Let the sibling be $x$, Yousef $5x$: $(x + 6) + (5x + 6) = 60 \Rightarrow 6x + 12 = 60 \Rightarrow 6x = 48 \Rightarrow x = 8$. Sibling is $8$, Yousef is $40$.

**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Algebra Essentials

Worker A can complete a job in 12 hours; Worker B can complete the same job in 4 hours. Working together, how long will it take them?

**a.** 8 **b.** 3 ✓

**c.** 2 **d.** 4

Combined rate $= \frac{1}{12} + \frac{1}{4} = \frac{1}{12} + \frac{3}{12} = \frac{4}{12}$ job/hour. Time $= 3$ hours.

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Geometry & Measurement

Worker A can complete a job in 8 hours; Worker B can complete the same job in 24 hours. Working together, how long will it take them?

**a.** 7 **b.** 6 ✓

**c.** 16 **d.** 5

Combined rate $= \frac{1}{8} + \frac{1}{24} = \frac{3}{24} + \frac{1}{24} = \frac{4}{24}$ job/hour. Time $= 6$ hours.

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**45** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Quantitative Comparisons

After spending half her money on a book, then 13 riyals on lunch, Huda has 18 riyals left. How much did she start with?

**a.** 72

**b.** 62 ✓

**c.** 18

**d.** 52

Let the start be $x$: $\frac{x}{2} - 13 = 18 \Rightarrow \frac{x}{2} = 31 \Rightarrow x = 62$ riyals.

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